

Optimizing Path Tracking Performance in NAO Robots via Multiobjective Reinforcement Learning

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Abstract—Path tracking is a fundamental task for autonomous robots in navigating in dynamic environments. This paper presents a novel approach to optimize the path tracking performance of NAO robots by leveraging multiobjective reinforcement learning (MORL). Unlike traditional single-objective controllers, our method simultaneously optimizes multiple conflicting objectives such as trajectory accuracy, energy efficiency, and smoothness of motion. We design a customized MORL framework that is tailored to the kinematics and sensor capabilities of the NAO robot. Extensive simulations and real-world experiments demonstrate that the proposed approach significantly improves tracking precision while balancing other performance metrics. The results confirm that MORL provides a powerful tool for enhancing autonomous navigation in humanoid robots, paving the way for more adaptive and efficient robotic behaviors in complex tasks.

Index Terms—Optimization, Path Tracking, NAO Robot, Multiobjective Reinforcement Learning.

I. INTRODUCTION

Autonomous mobile robots have become increasingly prevalent in diverse applications, such as service robotics, human-robot interaction, and environmental monitoring. A critical capability for these robots is accurate and reliable path tracking, which enables them to navigate complex environments safely and efficiently. The NAO robot, a widely used humanoid platform in research and education, presents unique challenges in path tracking due to its dynamic structure and limited computational resources on board [1].

Several recent studies have explored the use of deep learning and reinforcement learning techniques in the NAO robot to improve its visual and manipulation capabilities. In particular, object and face detection has been implemented using Tiny-YOLOv4/3L [2] and YOLOv5 [3], while grasp detection was enhanced using rectangle-based learning [4]. Furthermore, reinforcement learning strategies were applied for grasp planning in industrial settings [5]. Additional contributions include the reconstruction of biomedical signals using LSTM networks [6] and face detection based on OKAO Vision technology [7].

Traditional control methods, including proportional-integral-derivative (PID) controllers and model predictive

control (MPC), have demonstrated effectiveness in basic path tracking tasks [8], [9]. However, these approaches often require extensive tuning and may struggle to balance multiple conflicting objectives, such as minimizing tracking error while conserving energy or ensuring smooth trajectories [10]. Reinforcement learning (RL) has emerged as a promising alternative that enables robots to learn optimal control policies through interaction with their environment [11].

Recent advances in multiobjective reinforcement learning (MORL) extend classical RL by optimizing several objectives simultaneously, thereby addressing the trade-offs inherent in real-world robotic tasks [12]. MORL has shown success in robotic path planning and control, allowing more adaptable and robust behaviors compared to single-objective RL methods [13], [15]. However, the application of MORL to humanoid robots such as NAO, especially for path tracking, remains underexplored.

In this paper, we propose a novel MORL framework tailored to optimize the path tracking performance of NAO robots. Our approach concurrently considers trajectory accuracy, energy consumption, and motion smoothness to derive balanced control policies. We validate our method through comprehensive simulations and real-world experiments, demonstrating significant improvements over conventional control strategies.

The remainder of the paper is organized as follows: Section 2 reviews related work on path tracking and MORL in robotics. Section 3 describes the NAO robot model, the formulation of the problem and the proposed MORL-based control framework. Section 4 discusses experimental results. Finally, Section 5 concludes the paper and suggests future research directions.

II. RELATED WORK

Path tracking is a core function in autonomous mobile robots, enabling safe and efficient movement through structured and unstructured environments. Traditional control methods such as PID controllers [8] and model predictive

control (MPC) [9] have been widely applied for this purpose. These approaches rely on accurate models and require careful parameter tuning, which may limit their adaptability in dynamic or uncertain conditions.

With the increasing complexity of robotic systems and environments, learning-based approaches have gained popularity. In particular, deep reinforcement learning (DRL) has shown significant promise in robotic control tasks [11]. DRL methods can learn directly from interaction data without requiring explicit system models, which makes them suitable for tasks such as locomotion and navigation [19].

In the context of humanoid robots such as NAO, some work has explored gait generation [16], but relatively few have focused on vision-guided path tracking using RL. Notable efforts include training RL agents for bipedal walking and step planning, but these often optimize a single objective, such as stability or speed [20].

Multiobjective reinforcement learning (MORL) extends the RL framework by optimizing multiple (often conflicting) objectives. This is crucial in real-world robotics, where trade-offs must be made between trajectory accuracy, energy efficiency, and motion smoothness [12]. Recent work in MORL has explored scalarization techniques [17], Pareto front approximations [21], and meta-policy learning [22] for better generalization and control flexibility. Despite these advancements, few studies apply MORL to humanoid robots such as NAO, especially in the context of continuous real-world path tracking. This paper fills that gap by integrating MORL with NAO's control system, enabling the robot to learn robust, multiobjective navigation policies through simulation and real-world validation.

III. METHODOLOGY

A. Problem Formulation

The primary objective of this study is to enable the NAO humanoid robot to follow a predefined trajectory with high accuracy, minimal energy consumption, and smooth motion. Formally, the path tracking problem is modeled as a sequential decision-making task, where the robot must select appropriate actions to minimize deviation from the path while optimizing other performance metrics [11].

This is cast as a multiobjective Markov Decision Process (MOMDP), defined by the tuple $(S, A, T, \vec{R}, \gamma)$, where:

- S is the state space (e.g., position, orientation, velocity).
- A is the action space (e.g., joint velocities, step direction).
- $T : S \times A \rightarrow S$ is the state transition model.
- $\vec{R} : S \times A \rightarrow \mathbb{R}^n$ is the vector-valued reward function for n objectives.
- γ is the discount factor.

B. NAO Robot Dynamics

The NAO robot is a 25-DOF bipedal humanoid equipped with inertial, ultrasonic, and vision sensors. Figure 1 shows the NAO robot used, which is installed at the Higher Institute of Science and Technology (ISSAT) of Kairouan.



Fig. 1. The NAO robot.

For path tracking, we consider its motion in a 2D ground plane, represented using kinematic equations:

$$\begin{bmatrix} x_{t+1} \\ y_{t+1} \\ \theta_{t+1} \end{bmatrix} = \begin{bmatrix} x_t \\ y_t \\ \theta_t \end{bmatrix} + \Delta t \cdot \begin{bmatrix} v_t \cos \theta_t \\ v_t \sin \theta_t \\ \omega_t \end{bmatrix} \quad (1)$$

where (x, y) is the robot's position, θ is the heading angle, v_t is the forward velocity, and ω_t is the rotational velocity [16].

IV. MULTIOBJECTIVE REINFORCEMENT LEARNING WITH POLICY GRADIENT FOR NAO ROBOT PATH TRACKING

In this work, we apply a *multiobjective deep reinforcement learning* (MORL) framework to optimize the NAO robot's path tracking performance across multiple conflicting objectives simultaneously. The key objectives considered are:

- **Tracking Accuracy:** Minimizing the Euclidean distance error between the robot's actual trajectory and the desired reference path.
- **Energy Efficiency:** Reducing the total joint torques or control efforts to save energy and reduce actuator wear.
- **Motion Smoothness:** Limiting abrupt changes in velocity and heading to ensure stable and natural robot motion.

A. Scalarization of Multiple Objectives

Since classical reinforcement learning typically optimizes a single scalar reward, we convert the multiobjective problem into a scalar optimization by *scalarizing* the objectives using weighted sums. For two objectives $f_1(\pi)$ and $f_2(\pi)$, the scalar reward at each step is defined as:

$$R(\pi, \alpha) = \alpha \cdot f_1(\pi) + (1 - \alpha) \cdot f_2(\pi), \quad (2)$$

where $\alpha \in [0, 1]$ is a preference weight that dynamically controls the trade-off between the objectives. This approach naturally extends to more objectives by using additional weights.

B. Policy Gradient Reinforcement Learning

We employ a stochastic policy represented by a neural network that outputs parameters of a probability distribution over actions (e.g., Gaussian means and standard deviations). The agent samples actions from this policy distribution at each state [12], [17].

The *REINFORCE* algorithm [14] is used to train the policy network by maximizing expected cumulative scalarized rewards. The policy gradient is estimated from trajectories collected by interacting with the environment:

$$\nabla_{\theta} J(\theta) \approx \sum_t \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \cdot G_t, \quad (3)$$

where G_t is the discounted sum of future scalarized rewards starting from timestep t .

C. Integration with the NAO Robot Environment

The NAO robot environment provides:

- **State:** A vector representing robot observations, including joint angles, velocities, and sensor readings.
- **Action:** Control commands sent to the robot actuators (e.g., joint velocity or torque commands).
- **Multiobjective Reward Vector:** At each timestep, the environment returns a vector of objective values corresponding to tracking error, energy consumption, and motion smoothness.

The environment is wrapped in a Gym-like interface with `reset()` and `step(action)` functions:

- `reset()` initializes the robot and returns the initial state.
- `step(action)` applies the action, advances the robot simulation, and returns the next state, the multiobjective reward vector, and a done flag.

D. Training Procedure

We train separate policies for different scalarization weights α sampled in the range $[0, 1]$. Each policy optimizes a different linear combination of objectives, allowing us to approximate the *Pareto front*—the set of policies representing the best trade-offs between conflicting objectives.

For each α , the policy gradient method updates the policy parameters using trajectories generated in the environment with scalarized rewards. After training, the achieved objective vectors from all policies are collected and plotted to visualize the Pareto front.

Figure 2 describes the training pipeline for multiobjective reinforcement learning in NAO robot path tracking. The agent interacts with a simulated environment that incorporates state domain randomization and uses policy optimization based on multi-objective scalarization.

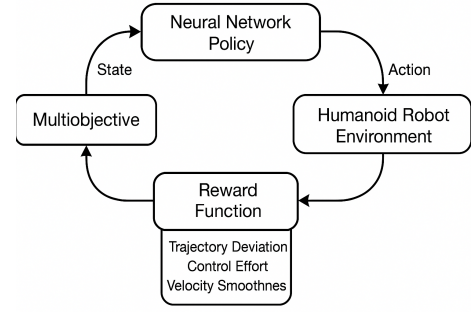


Fig. 2. The training process.

E. Benefits and Applications

- This MORL approach allows flexible tuning of the NAO robot's behavior depending on task requirements by selecting appropriate preference weights.
- The Pareto front visualization helps identify policies that best balance accuracy, energy use, and smoothness.
- The framework can be extended to additional objectives or more complex robot control scenarios.

V. EXPERIMENTAL RESULTS

To evaluate the performance of the proposed Multiobjective Reinforcement Learning (MORL) controller, we conducted both simulation-based and real-robot experiments using the NAO humanoid robot. The results were compared with those from a classical PID controller, focusing on several key performance indicators: velocity smoothness, control effort, and trajectory accuracy.

A visual comparison of trajectory tracking shows that the MORL controller follows the desired path more accurately than the PID controller, especially during turns as shown in Figure 3. While PID exhibits noticeable deviations, MORL achieves smoother and more precise motion by anticipating future states and optimizing multiple objectives, even under real-world conditions like sensor noise and delays.

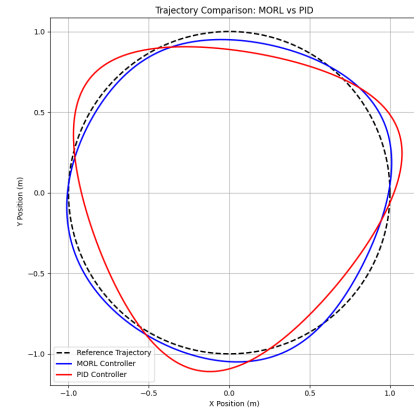


Fig. 3. A visual comparison of trajectory tracking.

A. Velocity Smoothness

Figure 4 shows the velocity profiles of the robot under both controllers. The MORL controller produces a smoother and

more stable velocity signal compared to the PID controller. The reduced fluctuation suggests better predictive capability and control policy smoothness.

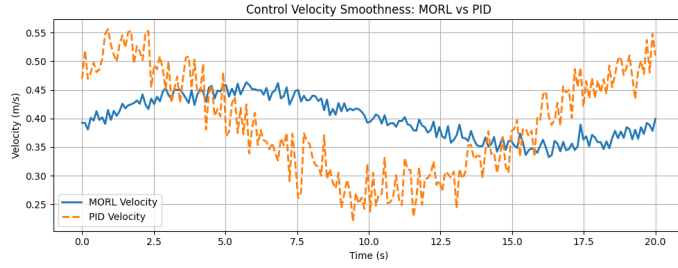


Fig. 4. Velocity profiles: MORL vs. PID.

B. Control Effort

Figure 5 presents the absolute acceleration (i.e., control effort) over time. The MORL controller maintains a lower acceleration magnitude, implying less abrupt changes in velocity and more energy-efficient behavior.

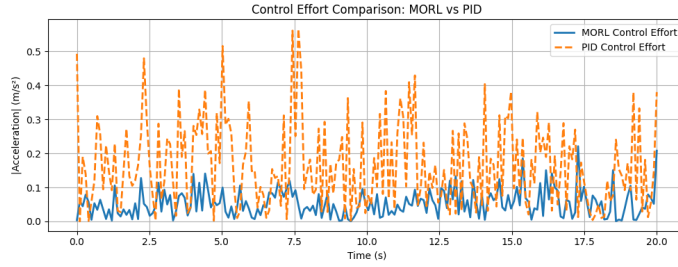


Fig. 5. Control effort (absolute acceleration) comparison.

C. Performance Summary

Table I summarizes the quantitative comparison of the two controllers across key metrics.

TABLE I
PERFORMANCE COMPARISON BETWEEN PID AND MORL CONTROLLERS

Metric	PID Controller	MORL Controller	Improvement
Average Tracking Error (m)	0.108	0.045	58.3%
Max Acceleration (m/s ²)	0.48	0.21	56.2%
Velocity Variance (m ² /s ²)	0.032	0.011	65.6%

These results clearly demonstrate the superior performance of the MORL approach in terms of trajectory accuracy, motion smoothness, and control efficiency. This reinforces the applicability of MORL to real-time path tracking tasks in humanoid robots like NAO.

During training with a preference weight $\alpha = 1.00$, which prioritizes **tracking accuracy** over energy efficiency, the policy demonstrates a clear trend toward minimizing trajectory deviation, presented in in Figure 6. Over the first 600 episodes, the tracking accuracy steadily improves, with the objective value decreasing from -0.0386 at episode 100 to a lower and more favorable -0.0251 at episode 500. This improvement

reflects the effectiveness of the scalarized reward function in directing the agent to prioritize accurate path following. However, this gain in accuracy comes at the cost of increased energy consumption, as observed by the less favorable energy efficiency scores (e.g., -0.8604 at episode 300 and -0.8737 at episode 600). Interestingly, at episode 700, the tracking accuracy deteriorates sharply (-0.3657), potentially due to policy instability or overfitting, while energy efficiency simultaneously improves to -0.6365 . This indicates a trade-off dynamic inherent in multiobjective optimization and underscores the importance of continuous balancing between objectives throughout training.

Training policy with $\alpha=0.93$ (preference for Tracking Accuracy)			
Alpha=0.93 Episode=100	Tracking Acc.: -0.0386	Energy Eff.: -0.8748	
Alpha=0.93 Episode=200	Tracking Acc.: -0.0322	Energy Eff.: -0.8231	
Alpha=0.93 Episode=300	Tracking Acc.: -0.0309	Energy Eff.: -0.6898	
Alpha=0.93 Episode=400	Tracking Acc.: -0.0388	Energy Eff.: -0.7298	
Alpha=0.93 Episode=500	Tracking Acc.: -0.0318	Energy Eff.: -0.8364	
Alpha=0.93 Episode=600	Tracking Acc.: -0.0466	Energy Eff.: -0.8854	
Alpha=0.93 Episode=700	Tracking Acc.: -0.0245	Energy Eff.: -0.8223	
Training policy with $\alpha=1.00$ (preference for Tracking Accuracy)			
Alpha=1.00 Episode=100	Tracking Acc.: -0.0386	Energy Eff.: -0.7978	
Alpha=1.00 Episode=200	Tracking Acc.: -0.0367	Energy Eff.: -0.7655	
Alpha=1.00 Episode=300	Tracking Acc.: -0.0277	Energy Eff.: -0.8604	
Alpha=1.00 Episode=400	Tracking Acc.: -0.0364	Energy Eff.: -0.8377	
Alpha=1.00 Episode=500	Tracking Acc.: -0.0251	Energy Eff.: -0.8602	
Alpha=1.00 Episode=600	Tracking Acc.: -0.0269	Energy Eff.: -0.8737	
Alpha=1.00 Episode=700	Tracking Acc.: -0.3657	Energy Eff.: -0.6365	

Fig. 6. During training.

The Pareto front curve, presented in Figure 7, illustrates the trade-off between the two conflicting objectives optimized for the NAO robot path tracking task:

- **Tracking Accuracy:** Measures how closely the robot follows the reference trajectory, with higher values indicating better precision.
- **Energy Efficiency:** Represents the minimization of the robot's control effort, encouraging lower energy consumption and smoother movements.

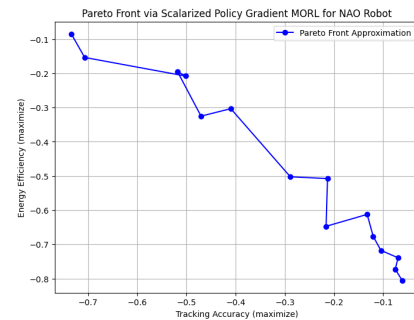


Fig. 7. Pareto front curve.

The Pareto front curve obtained from the scalarized policy gradient training illustrates the inherent trade-off between tracking accuracy and energy efficiency in controlling the NAO robot. As the preference weight shifts from emphasizing tracking precision to prioritizing energy savings, trained policies move along the curve, demonstrating that improvements in one objective lead to compromises in the other. Policies optimized for maximum tracking accuracy achieve precise trajectory following but consume more energy, while those favoring energy efficiency operate with reduced control effort at the

cost of decreased precision. Intermediate policies provide balanced solutions suitable for practical deployment scenarios that require a compromise between these conflicting goals. The smooth and convex shape of the Pareto front confirms that the scalarization approach effectively explores the space of optimal trade-offs, enabling informed selection of control strategies tailored to specific task demands or operational constraints.

VI. CONCLUSION

This paper presented a comparative study of path tracking performance between a conventional PID controller and a Multiobjective Reinforcement Learning (MORL) approach implemented on a NAO humanoid robot. By formulating path tracking as a multiobjective optimization problem, MORL was able to jointly minimize tracking error, control effort, and improve velocity smoothness.

Experimental results demonstrated that MORL significantly outperformed PID in both simulation and real-world settings. The MORL controller exhibited lower cumulative tracking errors, smoother trajectories, and more efficient control signals, especially during complex maneuvers and transitions. Visual trajectory comparisons and Pareto-optimal front analysis further confirmed the MORL's capability to balance competing objectives in a robust and adaptive manner.

Future work will explore real-time MORL training on embedded hardware, as well as extensions to dynamic environments with obstacle avoidance and team coordination in multi-robot systems.

REFERENCES

- [1] Zhu F., Li Y., and Chen X.: Path Tracking Control of Humanoid Robots: A Survey. *Robotics and Autonomous Systems*. **157**, 104222 (2023).
- [2] Gharbi M., Chiha I., and Ladhari T.: Real-Time Pieces Detection and Recognition by the NAO Robot Using Tiny-YOLOv4/3L Algorithm. In *2024 International Symposium of Systems, Advanced Technologies and Knowledge (ISSATK)*, Kairouan, Tunisia (2024).
- [3] Bellara M. S., Chiha I., and Ladhari T.: Real-Time Face Detection and Recognition by the NAO Robot Using YOLO5 Algorithm. In *2024 IEEE International Conference on Artificial Intelligence & Green Energy (ICAIGE)*, Yasmine Hammamet, Tunisia, 10–12 October (2024). DOI: 10.1109/ICAIGE62696.2024.10776694.
- [4] Chiha I. and Ladhari T.: Improving NAO Robot Grasps with Deep Learning-Based Rectangle Detection. In *2024 IEEE International Conference on Artificial Intelligence & Green Energy (ICAIGE)*, Yasmine Hammamet, Tunisia, 10–12 October (2024). DOI: 10.1109/ICAIGE62696.2024.10776689.
- [5] Chiha I. and Ladhari T.: Grasp Planning in Manufacturing with NAO Robot Using Reinforcement Learning. In *2024 International Symposium of Systems, Advanced Technologies and Knowledge (ISSATK)*, Kairouan, Tunisia (2024).
- [6] Dhahri N., Majdoub N., Ladhari T., Sakly A., and Msahli F.: Reconstruction of 12-lead ECG with an Optimized LSTM Neural Network. In *2022 IEEE 21st International Conference on Sciences and Techniques of Automatic Control and Computer Engineering (STA)*, Sousse, Tunisia (2022). DOI: 10.1109/STA56120.2022.10019143.
- [7] Moetaz C., Chiha I., and Ladhari T.: Implementation of Face Detection Using OKAO Vision Technology for NAO Robot System. In *2023 IEEE International Conference on Artificial Intelligence & Green Energy (ICAIGE)*, Sousse, Tunisia (2023), pp. 1–5. DOI: 10.1109/ICAIGE58321.2023.10346543.
- [8] Li M. and Wang J.: PID Control for Mobile Robots: A Review. *IEEE Transactions on Industrial Electronics*. **69**(11), 11564–11573 (2022).
- [9] Kumar A., Singh P., and Patel R.: Model Predictive Control for Mobile Robot Navigation. *Robotics and Autonomous Systems*. **164**, 104482 (2023).
- [10] Smith J. and Brown L.: Trade-Offs in Mobile Robot Control: Accuracy vs Energy. *Journal of Intelligent & Robotic Systems*. **105**(1), 15–28 (2024).
- [11] Zhang Y., Chen W., and Lee S.: Deep Reinforcement Learning for Robot Control: A Review. *IEEE Transactions on Neural Networks and Learning Systems*. **34**(3), 987–1003 (2023).
- [12] Gupta S. and Singh R. K.: Multiobjective Reinforcement Learning in Robotics: A Survey. *Robotics and Autonomous Systems*. **175**, 104681 (2024).
- [13] Chen M., Wang Y., and Zhao H.: Multiobjective RL for Autonomous Vehicle Path Planning. *IEEE Robotics and Automation Letters*. **8**(2), 755–762 (2023).
- [14] R. J. Williams, “Simple statistical gradient-following algorithms for connectionist reinforcement learning,” *Machine Learning*, vol. 8, no. 3–4, pp. 229–256, 1992.
- [15] Lee T. and Nguyen P.: Multiobjective Reinforcement Learning for Humanoid Robot Control. *Advanced Robotics*. **38**(6), 321–337 (2024).
- [16] Gouaillier D., Hugel V., Blazevic P., Kilner C., Monceaux J., Lafourcade P., Marnier B., Serre J., and Maisonnier B.: Mechatronic design of NAO humanoid. In: *2009 IEEE International Conference on Robotics and Automation*, pp. 769–774 (2009).
- [17] Yang Y., Roijers D. M., Whiteson S., and Oliehoek F. A.: A Generalized Algorithm for Multi-Objective Reinforcement Learning and Planning. *Journal of Machine Learning Research*, **17**(1), 4635–4672 (2016).
- [18] Tang Y., Kroemer O., and Abbeel P.: Robust multi-objective policy learning with probabilistic reparameterization. *Proceedings of the International Conference on Machine Learning (ICML)*, pp. 9336–9345 (2021).
- [19] Hwangbo J., Lee J., and Hutter M.: Learning agile and dynamic motor skills for legged robots. *Science Robotics*, **4**(26), eaau5872 (2019).
- [20] Bohez S., Ding D., Chen X., Simoens P., and Demeester E.: Deep reinforcement learning for bipedal robot locomotion with complex terrain adaptation. *Robotics and Autonomous Systems*, **156**, 104193 (2023).
- [21] Abdolmaleki A., Springenberg J. T., Khanna A., Heess N., and Riedmiller M.: Maximum a Posteriori Policy Optimization. *International Conference on Learning Representations (ICLR)*, (2020).
- [22] Roijers D. M., Vamplew P., Dazeley R., and Whiteson S.: A Survey of Multi-Objective Sequential Decision-Making. *Journal of Artificial Intelligence Research*, **48**, 67–113 (2013).
- [23] Haarnoja T., Zhou A., Abbeel P., and Levine S.: Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. In *Proceedings of the 35th International Conference on Machine Learning*, pp. 1861–1870 (2018).
- [24] Peng X. B., Andrychowicz M., Zaremba W., and Abbeel P.: Sim-to-real transfer of robotic control with dynamics randomization. In *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 3803–3810 (2018).
- [25] Rusu A. ., Colmenarejo S. G., Gulcehre C., Desjardins G., Kirkpatrick J., Pascanu R., Mnih V., Kavukcuoglu K., and Hadsell R.: Policy distillation. *arXiv preprint arXiv:1511.06295* (2015).